

1. Question

Write a Java program to demonstrate exception handling using `IllegalStateException` when an operation is performed on a closed `Scanner` object.

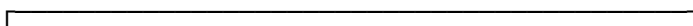
2. Steps for Execution

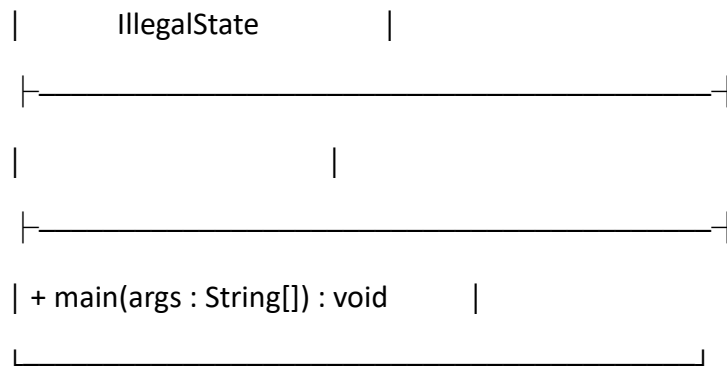
1. Open a Java IDE or command prompt.
 2. Create a package named `javapractice`.
 3. Create a class named `IllegalState`.
 4. Create a `Scanner` object to read input.
 5. Close the `Scanner` using `sc.close()`.
 6. Try to read input using `sc.nextLine()`.
 7. Since the `Scanner` is already closed, an `IllegalStateException` occurs.
 8. Catch the exception using the `catch` block.
 9. Display "Scanner is already closed".
 10. Compile and execute the program.
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3. Algorithm

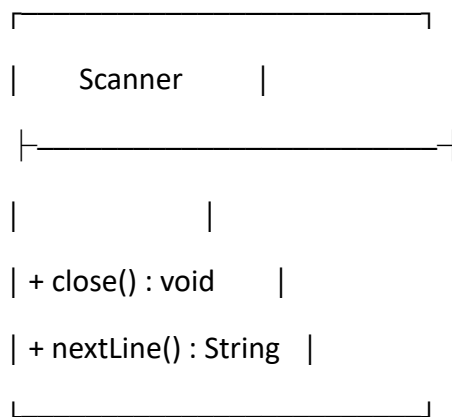
1. **Start**
 2. Create a `Scanner` object.
 3. Close the `Scanner`.
 4. Try to read a line using `nextLine()`.
 5. Since the `Scanner` is closed, generate `IllegalStateException`.
 6. Catch the exception using the `catch` block.
 7. Display "Scanner is already closed".
 8. **Stop**
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4. UML Class Diagram

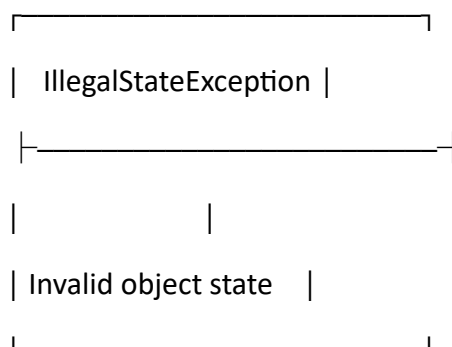




| creates / uses



| operation after close



5. Code

```
package javapractice;
```

```
public class IllegalState {  
  
    public static void main(String[] args) {  
        try {  
            java.util.Scanner sc = new java.util.Scanner(System.in);  
  
            sc.close();  
  
            System.out.println(sc.nextLine());  
        }  
        catch (IllegalStateException e) {  
            System.out.println("Scanner is already closed");  
        }  
    }  
}
```

6. Output

Scanner is already closed

7. Result

Thus, the Java program successfully demonstrates exception handling using `IllegalStateException` when an operation is performed on an already closed Scanner object.